

BLUESTOCK FINTECH PVT. LTD.

Capstone Project Report

Mutual Fund Analytics Platform

An End-to-End Data Analytics Pipeline for Indian Mutual Fund Market Intelligence

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I am also grateful to the Association of Mutual Funds in India (AMFI) and the MFAPI open data platform for making mutual fund data publicly accessible, which formed the backbone of the data ingestion and analytics pipeline developed in this project.

Finally, I extend my gratitude to my peers, faculty members, and family for their continuous support and encouragement throughout the project.

Manoj Seervi

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Executive Summary

The Mutual Fund Analytics Platform is an end-to-end data analytics solution built to provide deep insights into the Indian mutual fund industry using publicly available data from AMFI and MFAPI. Designed and implemented over six intensive days, the platform spans the complete analytics value chain — from automated data ingestion and cleaning through to risk modeling, investor cohort analysis, and interactive Power BI dashboards.

The platform integrates ten distinct datasets including Fund Master data, NAV History, AUM by Fund House, Monthly SIP Inflows, Category Inflows, Industry Folio Counts, Scheme Performance, Investor Transactions, Portfolio Holdings, and Benchmark Indices. All data is stored in a structured SQLite relational database comprising six tables — two-dimension tables and four fact tables — ensuring efficient querying and analytical flexibility.

Performance analytics cover a comprehensive set of financial metrics: Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown, enabling fund managers and investors to evaluate risk-adjusted returns objectively. Advanced risk analytics further extend the platform with Historical Value at Risk (VaR), Conditional VaR (CVaR), Rolling Sharpe Ratio, and Sector Concentration using the Herfindahl-Hirschman Index (HHI).

The investor analytics module conducts cohort analysis and SIP continuity studies to identify at-risk investors, while the rule-based Recommendation Engine classifies funds into Low, Moderate, and High-risk categories to support investor suitability matching. The five-page Power BI dashboard provides a unified visual interface covering Industry Overview, Fund Performance, NAV vs Benchmark, Investor Analytics, and SIP & Market Trends.

The platform demonstrates the practical application of data engineering, financial analytics, and business intelligence in a real-world fintech context, and serves as a production-ready prototype for mutual fund intelligence at Bluestock Fintech Pvt. Ltd.

The project demonstrates how modern data analytics, financial modeling, and business intelligence tools can be integrated to support informed investment decision-making in the mutual fund industry.

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1. Introduction

The Indian mutual fund industry has witnessed exponential growth over the past decade, with Assets Under Management (AUM) crossing INR 50 lakh crore and a rapidly expanding retail investor base. The proliferation of fund schemes, coupled with varying risk-return profiles and market dynamics, has created a pressing need for robust analytical tools that can help investors, fund managers, and regulators make informed decisions.

This capstone project, developed at Bluestock Fintech Pvt. Ltd., addresses this need by building a Mutual Fund Analytics Platform — a comprehensive, data-driven solution that automates the ingestion, processing, analysis, and visualization of mutual fund data sourced from AMFI and MFAPI. The platform is designed to serve as an internal analytics backbone for fintech applications, offering insights at both the macro industry level and the granular fund-scheme level.

The project adopts an agile, day-by-day sprint structure, simulating a real-world fintech product development lifecycle. Each day introduces a new layer of analytical depth — from raw data collection and cleaning on Day 1 and 2, through exploratory analysis and performance analytics on Day 3 and 4, to advanced risk modeling, investor analytics, and dashboard delivery on Day 5 and 6. The result is a fully functional analytics platform built entirely on open-source technologies and publicly available data.

2. Problem Statement

Despite the availability of vast amounts of mutual fund data through AMFI and other regulatory disclosures, the Indian fintech ecosystem lacks accessible, integrated tools that consolidate and analyze this data in a structured manner. Fund performance comparisons are often made in isolation, without considering risk-adjusted metrics, benchmark deviations, or investor behavioural patterns.

Key challenges include: (a) fragmented data across multiple sources with inconsistent formats; (b) the absence of a unified database for fund performance, investor transactions, and benchmark indices; (c) limited tools for quantifying downside risk beyond simple return metrics; (d) no systematic mechanism to identify at-risk SIP investors or provide personalized fund recommendations; and (e) a lack of interactive, multi-dimensional dashboards that communicate fund analytics to non-technical stakeholders.

This project directly addresses these gaps by building a production-ready analytics pipeline that ingests, cleans, stores, analyzes, and visualizes Indian mutual fund data in a structured, scalable, and interpretable manner.

3. Objectives

The Mutual Fund Analytics Platform is built with the following specific objectives:

- Design and implement an automated data ingestion pipeline integrating AMFI and MFAPI data sources, covering ten distinct datasets relevant to the Indian mutual fund market.
- Clean, preprocess, and normalize raw datasets to ensure data quality, consistency, and analytical readiness.
- Design and populate a normalized SQLite relational database with clearly defined dimension and fact tables to support structured analytical queries.
- Conduct comprehensive Exploratory Data Analysis (EDA) to uncover patterns in fund performance, investor demographics, SIP trends, and sector allocations.
- Implement standard financial performance metrics — CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown — for objective fund evaluation.
- Develop advanced risk analytics including Historical VaR, Conditional VaR (CVaR), Rolling Sharpe Ratio, and Sector Concentration Index (HHI).
- Conduct investor cohort analysis and SIP continuity studies to identify at-risk investors and investment behavior patterns.
- Build a risk-based Fund Recommendation Engine that matches investors with appropriate fund categories based on their risk profile.
- Design and deploy a five-page interactive Power BI dashboard that presents all analytics in a visual, accessible, and business-ready format.

4. Mutual Fund Industry Overview

Mutual funds in India are regulated by the Securities and Exchange Board of India (SEBI) and are managed by Asset Management Companies (AMCs). They pool capital from investors to invest in diversified portfolios of equity, debt, hybrid, and other instruments. Key industry concepts relevant to this project include:

Term	Description	Category
NAV (Net Asset Value)	The per-unit market value of a fund's assets minus liabilities. Updated daily.	Core pricing metric
AUM (Assets Under Management)	Total market value of assets a fund house manages on behalf of investors.	Industry size indicator
SIP (Systematic Investment Plan)	A method of investing a fixed amount periodically in a mutual fund scheme.	Retail investor tool
Folio	A unique account number assigned to an investor in a fund house.	Investor count proxy
CAGR	Compound Annual Growth Rate — measures annualized return over a period.	Performance metric

Sharpe Ratio	Risk-adjusted return relative to a risk-free rate, normalized by volatility.	Risk-return metric
Alpha / Beta	Alpha: excess return over benchmark. Beta: fund sensitivity to market movements.	Factor analytics
VaR / CVaR	Value at Risk / Conditional VaR — measures potential downside loss at confidence levels.	Risk metrics
HHI	Herfindahl-Hirschman Index — measures sector concentration in a portfolio.	Concentration risk

The Indian mutual fund market is categorized by SEBI into Equity, Debt, Hybrid, Solution-Oriented, and other schemes, with further sub-categories based on investment strategy, duration, and risk profile. This classification framework serves as the foundation for fund categorization, risk profiling, performance comparison, and recommendation generation within the platform.

5. Data Sources

The platform draws upon two primary publicly available data sources for the Indian mutual fund ecosystem:

5.1 AMFI (Association of Mutual Funds in India)

AMFI is the industry body for mutual funds in India and publishes regulatory data including fund master data, NAV history, AUM reports, SIP inflow statistics, category-wise inflow data, industry folio counts, and investor demographics. Data is accessible via AMFI's official website and FTP endpoints in structured text and CSV formats.

5.2 MFAPI (Mutual Fund API)

MFAPI (<https://www.mfapi.in>) is an open-source, free REST API that provides historical NAV data for all AMFI-registered mutual fund schemes in JSON format. The API supports scheme-level NAV history retrieval by scheme code, making it ideal for automated batch ingestion. Two custom Python scripts — `data_ingestion.py` and `live_nav_fetch.py` — were developed to interact with these sources programmatically.

6. Dataset Description

Ten distinct datasets were collected, cleaned, and integrated into the analytics platform. The table below summarizes each dataset:

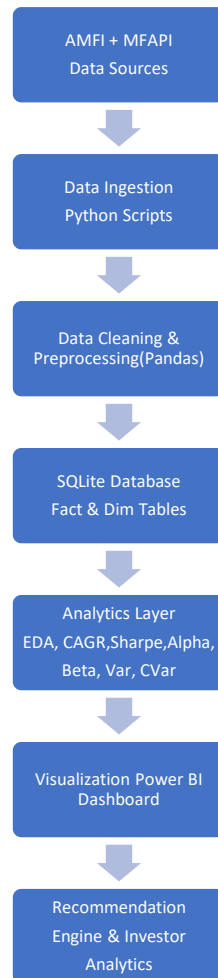
Dataset	Description	Source
Fund Master	Master list of all registered mutual fund schemes with AMC details, category, and scheme type.	AMFI
NAV History	Daily NAV values for each fund scheme over the historical period.	MFAPI
AUM by Fund House	Monthly AUM figures for each Asset Management Company.	AMFI
Monthly SIP Inflows	Aggregate monthly SIP contribution amounts and folio counts.	AMFI
Category Inflows	Net inflows by fund category (Equity, Debt, Hybrid, etc.) on a monthly basis.	AMFI
Industry Folio Count	Total number of investor folios across the industry by month.	AMFI
Scheme Performance	Point-to-point returns for 1M, 3M, 6M, 1Y, 3Y, 5Y periods for each scheme.	AMFI
Investor Transactions	Simulated/anonymized investor transaction records including SIP, Lumpsum, and Redemptions.	Synthetic + AMFI

Portfolio Holdings	Scheme-level sector and stock allocation data.	AMFI Disclosures
Benchmark Indices	Historical index values for Nifty 50, Sensex, and other benchmarks.	NSE/BSE Public Data

Together, these datasets provide a comprehensive view of the mutual fund ecosystem, covering fund characteristics, performance history, investor behavior, market benchmarks, portfolio composition, and industry-wide trends.

7. System Architecture

The platform is structured as a layered analytics pipeline with clearly defined stages, each building upon the output of the previous. The architecture follows an Extract-Transform-Load (ETL) paradigm extended with analytics and visualization layers.



The Mutual Fund Analytics Platform follows a layered architecture designed to transform raw mutual fund data into actionable business insights. Data is collected from AMFI and MFAPI through automated Python-based ingestion scripts and subsequently cleaned and standardized using Pandas-based preprocessing techniques. The processed data is stored in a structured SQLite database comprising dimension and fact tables to support efficient querying and analysis.

The analytics layer computes key performance and risk metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, CVaR, and Rolling Sharpe Ratio. Results are presented through interactive Power BI dashboards that provide industry-level and fund-level insights. The final intelligence layer includes investor cohort analysis, SIP continuity analysis, and a risk-based fund recommendation engine, enabling informed investment decision-making...

8. Data Ingestion Process

Data ingestion formed the foundation of the analytics pipeline and was responsible for collecting mutual fund data from multiple public sources. Two primary Python scripts were developed to automate the collection of mutual fund data from external sources:

8.1 data_ingestion.py

This script handles the batch download of static and semi-static datasets from AMFI's public endpoints, including Fund Master data, AUM reports, SIP inflow statistics, category inflow data, folio count reports, and scheme performance data. The script uses the requests and pandas libraries to fetch, parse, and store data as CSV files in the project's data directory.

8.2 live_nav_fetch.py

This script interfaces with the MFAPI REST API to fetch historical NAV data on a per-scheme basis. It iterates over the list of scheme codes from the Fund Master dataset and retrieves the complete NAV history for each scheme. JSON responses are parsed and consolidated into a single NAV history Data Frame, then written to CSV.

The project repository was organized into dedicated folders for raw data, processed datasets, database storage, notebooks, reports, and Python scripts. A total of ten datasets were successfully ingested, covering fund metadata, NAV history, performance metrics, investor transactions, portfolio holdings, and benchmark indices.

9. Data Cleaning & Preprocessing

The data cleaning and preprocessing phase focused on improving data quality, consistency, and reliability before loading datasets into the analytics database.— a critical step to ensure analytical accuracy and database integrity. The following cleaning operations were applied to the primary datasets:

9.1 NAV History Cleaning

The raw NAV dataset contained missing values for non-trading days, duplicate scheme entries, and inconsistent date formats. Missing NAVs were forward-filled to maintain continuity for rolling calculations. Date columns were parsed to a uniform datetime format. NAV values were validated against expected numeric ranges, and outliers caused by data entry errors were flagged and corrected.

9.2 Investor Transactions Cleaning

The investor transaction dataset included records with null transaction amounts, unrecognized transaction type codes, and mismatched investor IDs. Records with critical null values were dropped; transaction types were normalized to a standard taxonomy (SIP, Lumpsum, Redemption, Switch). Investor age and state fields were validated and standardized.

9.3 Scheme Performance Cleaning

Return values for various periods were stored as strings with percentage symbols and parentheses for negative values. These were converted to numeric floats. Schemes with incomplete return histories were retained with null flags for transparency. Category and sub-category labels were harmonized to match SEBI's official scheme classification.

Cleaning Operation	Approach
Missing Value Treatment	Missing values handled using appropriate imputation and validation techniques
Date Standardization	All date columns converted to ISO 8601 (YYYY-MM-DD) format
Data Type Conversion	Return percentages converted from strings to float64
Duplicate Removal	Deduplication applied on scheme code + date composite keys
Outlier Handling	Statistical bounds used to flag NAV anomalies; manual review applied
Category Harmonization	Scheme categories aligned to SEBI master classification taxonomy

The cleaning process significantly improved dataset consistency and ensured that downstream analytics, risk calculations, and dashboard visualizations were based on reliable and standardized data.

10. Database Design (SQLite)

A normalized relational database was designed and implemented in SQLite to serve as the central data repository for the analytics platform. The database follows a dimensional modeling approach using dimension and fact tables to support efficient querying, reporting, and analytical computations..

10.1 Database Schema

Table	Type	Key Columns
dim_fund	Dimension	Scheme code, scheme name, AMC name, category, sub-category, risk grade, benchmark, expense ratio
dim_date	Dimension	Date key, day, month, quarter, year, financial year, is_trading_day flag
fact_nav	Fact	Scheme code (FK), date key (FK), NAV value, daily return, 52W high, 52W low
fact_transactions	Fact	Transaction ID, investor ID, scheme code (FK), date key (FK), transaction type, amount, units
fact_performance	Fact	Scheme code (FK), period, return value, rank, category average, benchmark return
fact_aum	Fact	AMC name, date key (FK), AUM value, category, SIP AUM, no. of folios

10.2 Key Analytical SQL Queries

A set of analytical SQL queries was developed and documented in the data dictionary. These include: top-performing schemes by CAGR; category-wise AUM growth trends; month-on-month SIP inflow analysis; fund house market share by AUM; investor age-group distribution; and scheme-level transaction volume analysis. The queries form the basis for Power BI data model relationships and DAX measures.

10.3 Data Dictionary

A comprehensive data dictionary was created documenting every table, column, data type, allowed values, and business definition. This ensures consistency across analytics scripts and dashboard development, and serves as documentation for future platform enhancements.

11. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to identify patterns, trends, anomalies, and relationships within the collected mutual fund datasets. to uncover patterns, trends, and anomalies across the ten datasets. All visualizations were generated using Matplotlib and Seaborn, and saved to the /visualizations directory. The EDA results informed the selection of features and metrics for subsequent analytical modules.

11.1 NAV Trend Analysis

Time-series plots of NAV values for selected funds across categories revealed clear bull and bear market cycles, with equity funds showing higher volatility and greater long-term appreciation compared to debt funds.

11.2 AUM & Industry Analysis

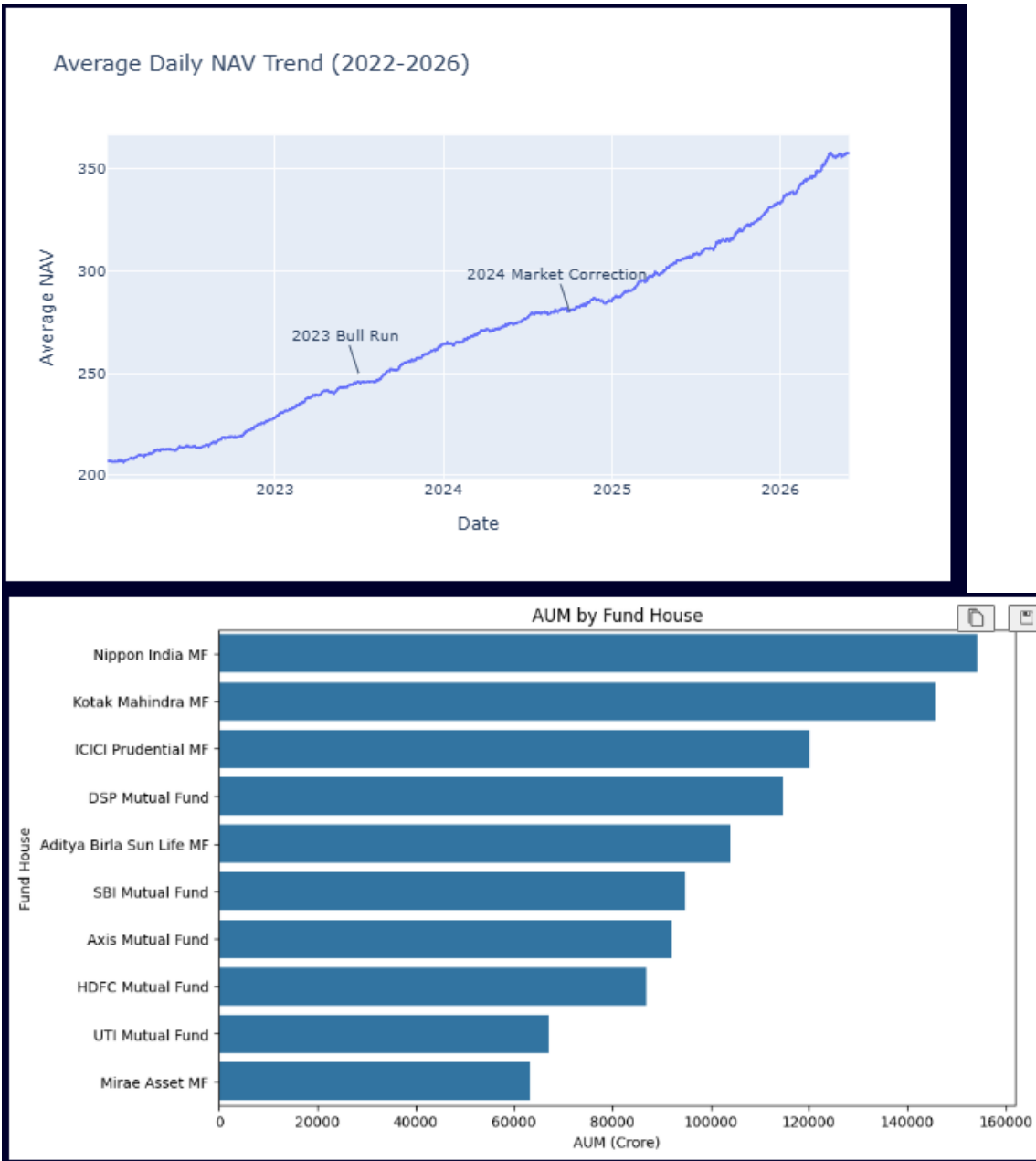
The AUM by Fund House analysis showed a highly concentrated market, with the top five fund houses (SBI Mutual Fund, HDFC Mutual Fund, ICICI Prudential, Nippon India, Kotak Mahindra) commanding over 60% of total industry AUM. The Folio Growth chart illustrated a consistent upward trend, reflecting growing retail participation in mutual funds post the Jan Dhan and PMJDY financial inclusion drives.

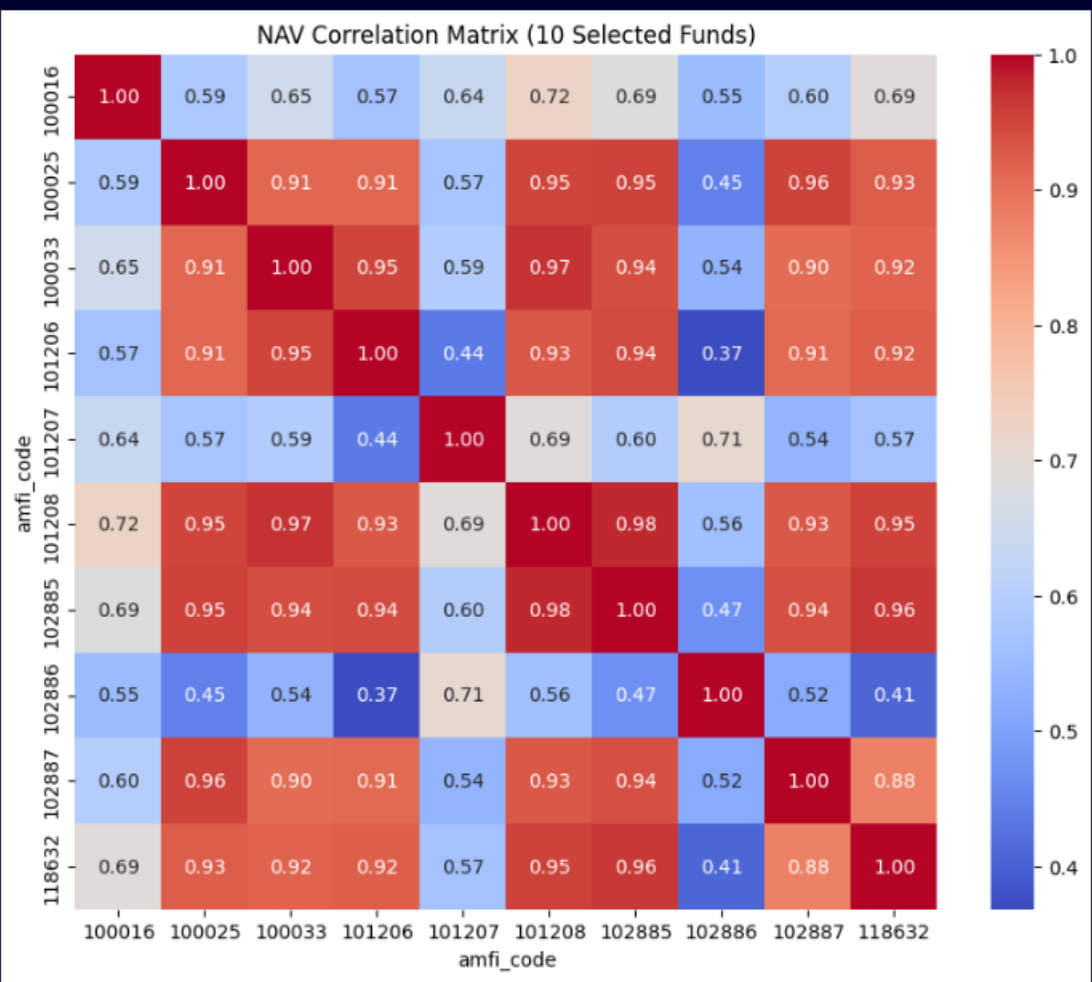
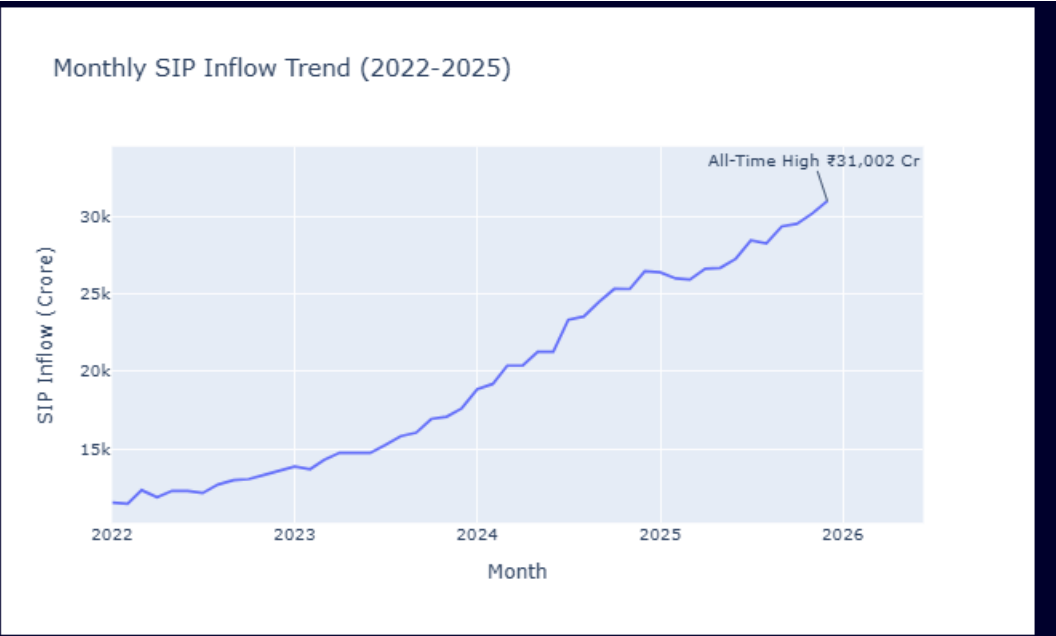
11.3 Investor Demographics

The age distribution analysis revealed a bimodal pattern — a younger cohort (25–35 years) representing new SIP investors and an older cohort (45–60 years) representing legacy lumpsum investors. Gender distribution showed a gradual increase in female investor participation. The state distribution highlighted the T30-B30 divide: investors from top 30 cities (T30) dominate AUM while bottom 30 cities (B30) show higher folio counts relative to AUM, indicating smaller ticket sizes.

11.4 Portfolio & Sector Analysis

The sector allocation analysis using stacked bar charts revealed sector concentration in Financial Services, Information Technology, and Consumer Discretionary across large-cap equity funds. The correlation matrix of fund returns identified clusters of highly correlated funds within the same category — useful for portfolio diversification analysis. The Risk Category distribution confirmed SEBI's mandated risk-o-meter classification, with "Very High" risk concentrated in small-cap and sectoral funds.





12. Fund Performance Analytics

This phase of the project focused on evaluating mutual fund performance using industry-standard financial metrics. Metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were computed using Python (Pandas and NumPy) and consolidated into the fund_scorecard dataset for further analytics and dashboard reporting.

12.1 Compound Annual Growth Rate (CAGR)

CAGR measures the annualized rate of return of a fund over a specified period, assuming reinvestment of returns. It provides a smoothed representation of growth, independent of interim volatility.

$$CAGR = [(Ending\ NAV / Beginning\ NAV)^{(1 / Number\ of\ Years)}] - 1$$

12.2 Sharpe Ratio

The Sharpe Ratio quantifies the excess return earned per unit of total risk (standard deviation), providing a risk-adjusted performance measure. A higher Sharpe Ratio indicates superior risk-adjusted performance. The risk-free rate used in this project is the prevailing 91-day T-Bill rate.

$$Sharpe\ Ratio = (Portfolio\ Return - Risk-Free\ Rate) / Standard\ Deviation\ of\ Returns$$

12.3 Sortino Ratio

The Sortino Ratio is a refinement of the Sharpe Ratio that penalizes only downside deviation (negative return volatility), rather than total standard deviation. It is more appropriate for non-symmetric return distributions common in equity funds.

$$Sortino\ Ratio = (Portfolio\ Return - Risk-Free\ Rate) / Downside\ Deviation$$

12.4 Alpha and Beta

Alpha measures the excess return of a fund relative to its benchmark, after accounting for systematic risk. A positive alpha indicates outperformance. Beta measures the fund's sensitivity to market movements — a beta greater than 1 indicates higher volatility than the market, while a beta less than 1 indicates lower sensitivity.

$$Alpha = Fund\ Return - [Risk-Free\ Rate + Beta * (Benchmark\ Return - Risk-Free\ Rate)]$$

$$Beta = Covariance\ (Fund\ Returns, Benchmark\ Returns) / Variance\ (Benchmark\ Returns)$$

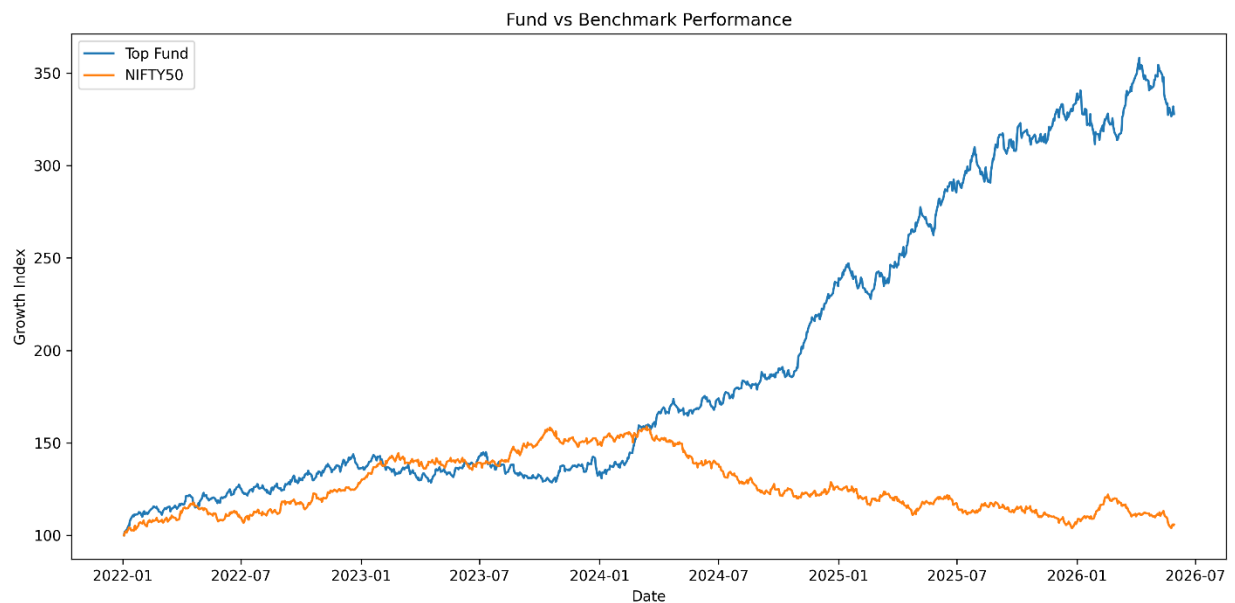
12.5 Maximum Drawdown

Maximum Drawdown (MDD) measures the maximum peak-to-trough decline in a fund's NAV over the analysis period. It represents the worst-case loss an investor could have experienced, making it a critical risk metric for capital preservation.

$$\text{Maximum Drawdown} = (\text{Trough NAV} - \text{Peak NAV}) / \text{Peak NAV}$$

12.6 Fund Scorecard

A consolidated Fund Scorecard was generated combining CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown for each scheme. Funds were ranked within their categories to identify top performers. The scorecard was exported as fund_scorecard.csv and used as the primary data source for the Power BI Fund Performance page.



13. Advanced Risk Analytics

Day 6 extended the analytics framework with advanced risk modeling techniques, providing a more nuanced assessment of downside exposure and portfolio concentration risk.

13.1 Value at Risk (VaR)

Historical VaR at the 95% confidence level was computed for each fund by ranking the daily return distribution and identifying the 5th percentile return. VaR provides a probabilistic estimate of the maximum expected loss over a one-day horizon under normal market conditions.

$$VaR (95\%) = 5th \text{ Percentile of Historical Daily Returns}$$

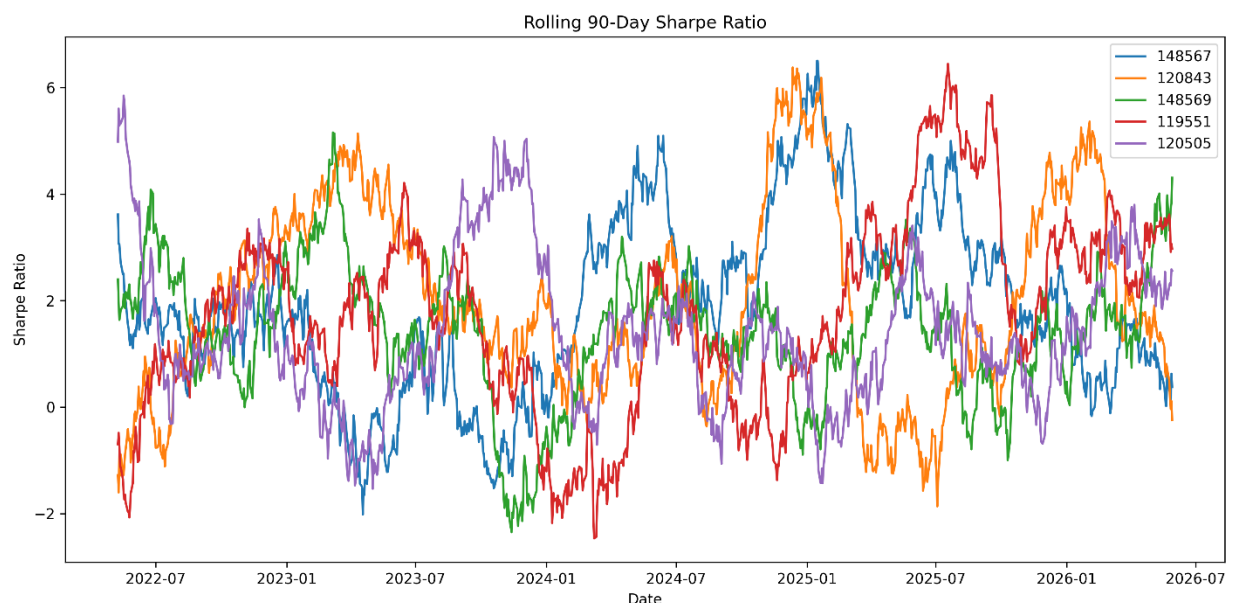
13.2 Conditional Value at Risk (CVaR)

CVaR, also known as Expected Shortfall (ES), extends VaR by computing the average of all losses that exceed the VaR threshold. It captures tail risk more comprehensively and is particularly relevant for stress testing and regulatory capital estimation.

$$CVaR (95\%) = \text{Mean of Daily Returns below VaR (95\%) threshold}$$

13.3 Rolling 90-Day Sharpe Ratio

To assess the consistency of a fund's risk-adjusted performance over time, a rolling 90-day Sharpe Ratio was computed using a sliding window approach. This metric identifies periods of performance degradation or improvement and helps investors understand whether a fund's strong Sharpe Ratio is sustained or episodic.



13.4 Sector Concentration Index (HHI)

The Herfindahl-Hirschman Index (HHI) was applied to measure portfolio sector concentration. A lower HHI indicates a well-diversified portfolio, while a higher HHI signals concentration risk. The metric is computed using the squared sum of sector weight proportions:

$$HHI = \sum (Sector\ Weight_i)^2 \quad \text{for all sectors } i \text{ in the portfolio}$$

An HHI below 0.10 indicates high diversification, 0.10–0.25 indicates moderate concentration, and above 0.25 indicates high concentration. The results were used to flag concentrated funds in the recommendation engine.

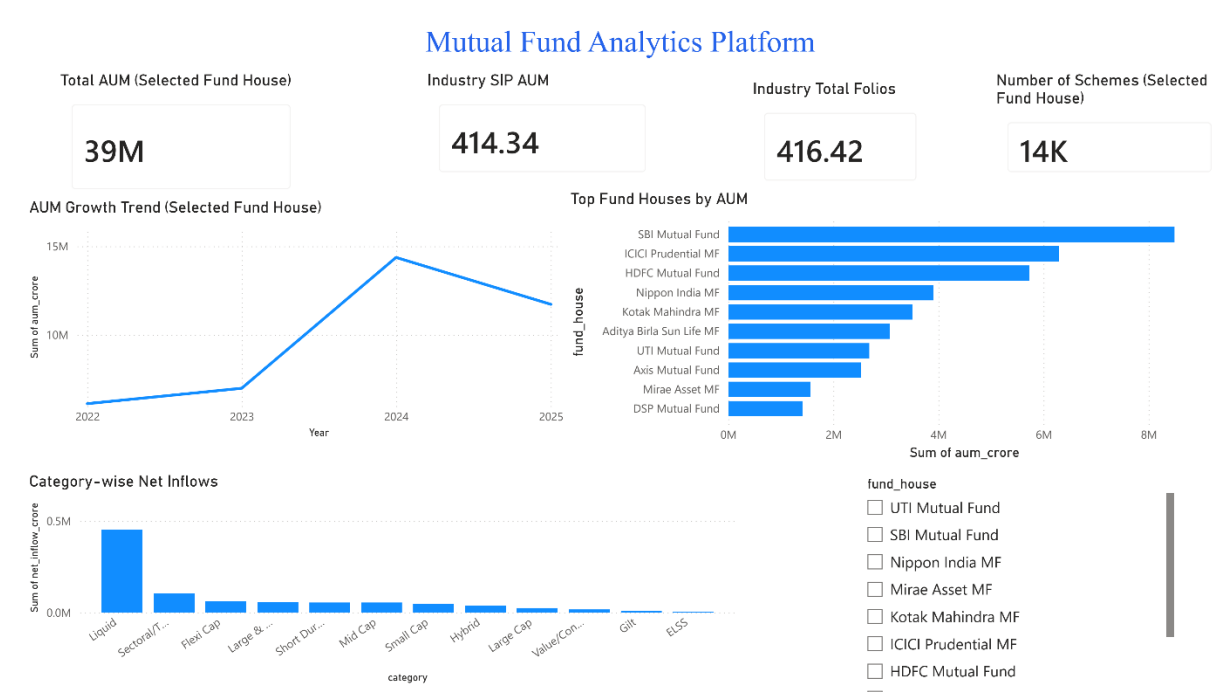
Risk Metric	Purpose
Historical VaR (95%)	Estimates maximum daily loss under normal market conditions (95% confidence)
Conditional VaR (CVaR)	Computes average loss in the worst 5% of scenarios — captures tail risk
Rolling Sharpe Ratio	90-day sliding window Sharpe — measures consistency of risk-adjusted returns
HHI (Sector Concentration)	Measures portfolio diversification across sectors; flags concentration risk

14. Power BI Dashboard Design

A comprehensive five-page Power BI dashboard was developed on Day 5, providing an interactive, stakeholder-ready interface for all analytics outputs. The dashboard connects directly to the SQLite database via ODBC and to the exported CSV files (fund_scorecard.csv, alpha_beta.csv). Cross-page slicers for Fund House and Category enable dynamic filtering across all pages.

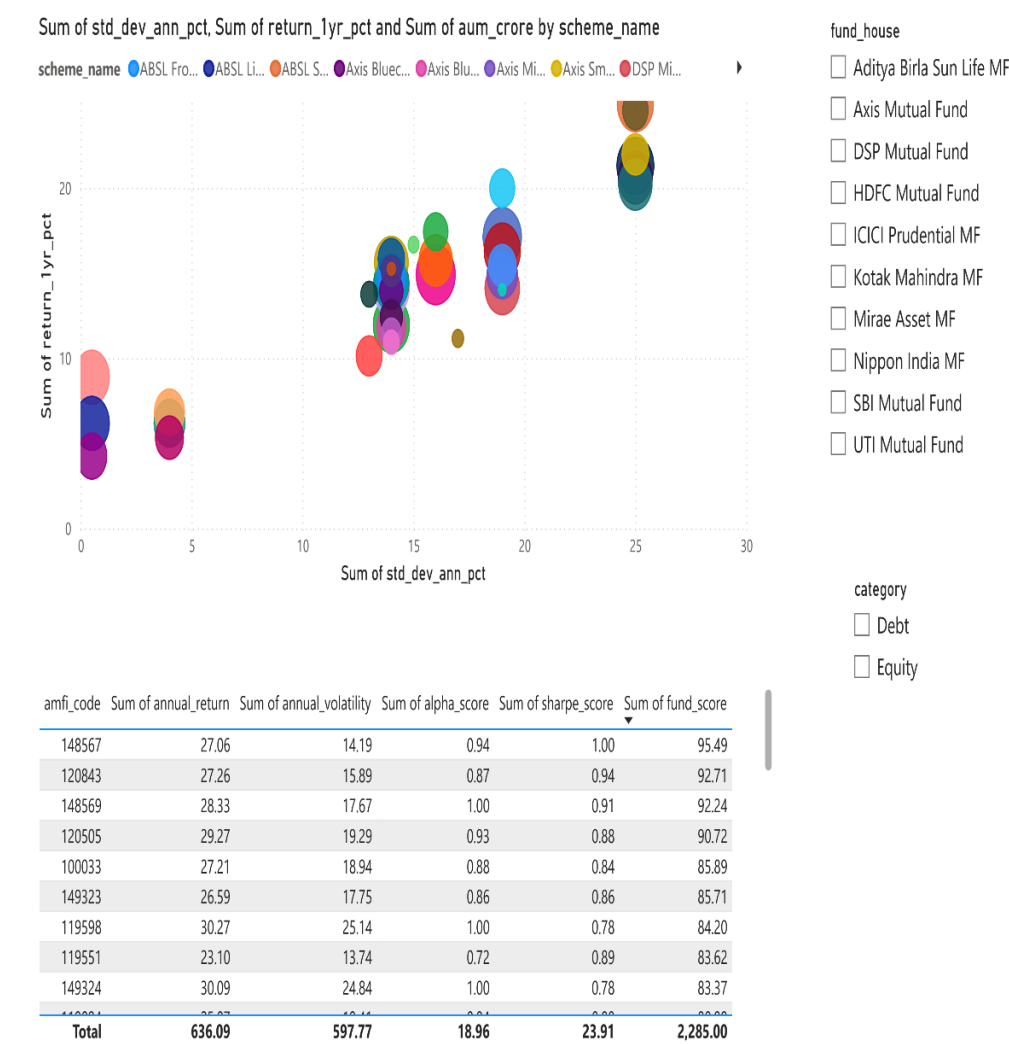
14.1 Page 1 — Industry Overview

The Industry Overview page provides a macro-level snapshot of the Indian mutual fund industry. Key KPI cards display Total AUM, SIP AUM, Total Industry Folios, and Number of Active Schemes. An AUM Growth Trend line chart tracks monthly AUM evolution over the past three years. A horizontal bar chart ranks Top Fund Houses by AUM, while a donut chart breaks down Category Inflows (Equity, Debt, Hybrid, Others).



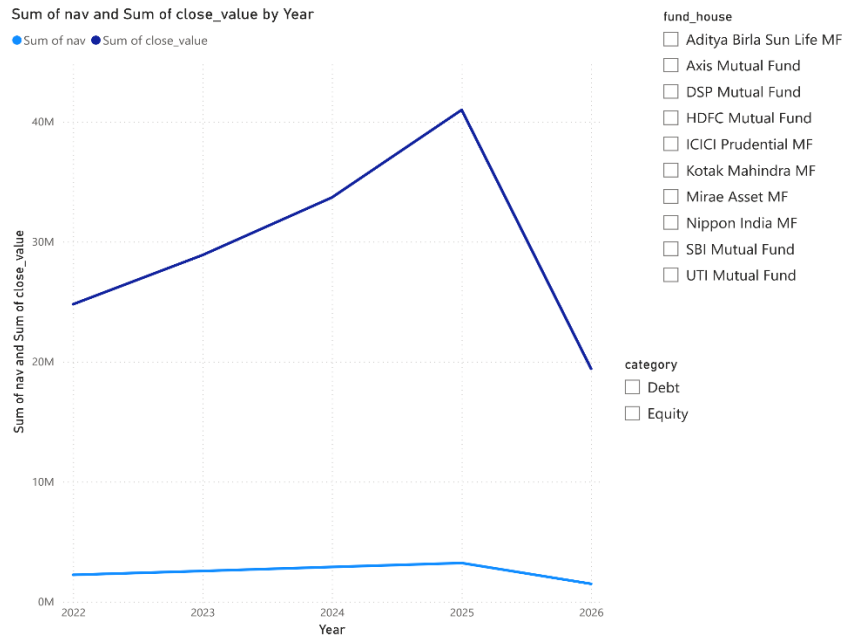
14.2 Page 2 — Fund Performance

The Fund Performance page centers on a scatter plot of Return (Y-axis) vs Risk / Standard Deviation (X-axis), with bubble size representing AUM. This visual allows instant identification of high-return, low-risk funds (top-left quadrant). The Fund Scorecard table below shows CAGR, Sharpe Ratio, Alpha, Beta, and Maximum Drawdown per scheme, with conditional formatting. Fund House and Category slicers filter all visuals simultaneously.



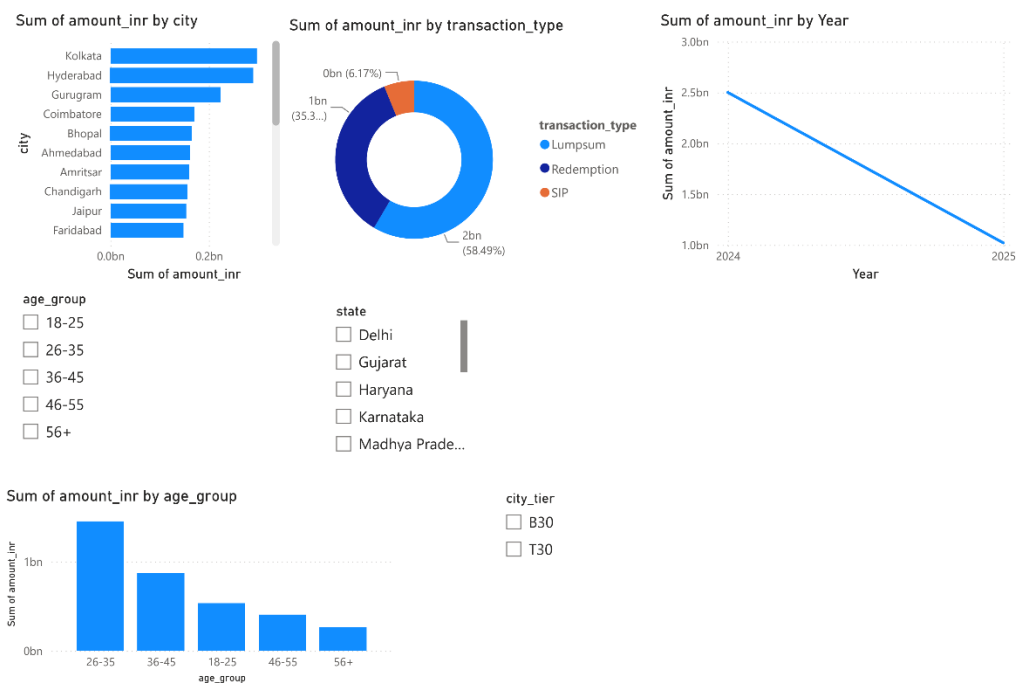
14.3 Page 3 — NAV vs Benchmark

This page overlays the NAV trend of a selected fund against its benchmark index (e.g., Nifty 50), enabling visual comparison of relative performance. A dual-axis line chart plots both time series. Period selectors (1M, 3M, 6M, 1Y, 3Y, 5Y) allow dynamic time horizon adjustments. Alpha and Beta KPIs are displayed as summary cards.



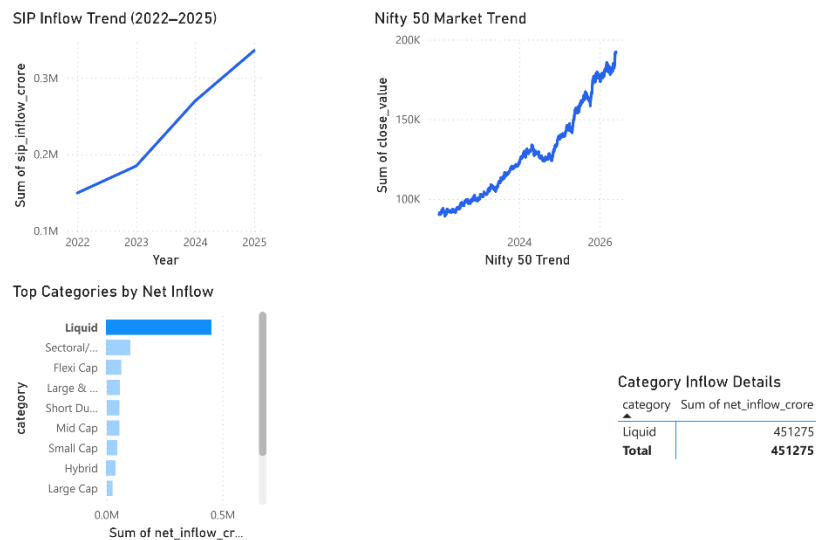
14.4 Page 4 — Investor Analytics

The Investor Analytics page visualizes the investor base composition. Stacked bar charts show Transaction Volumes by Type (SIP, Lumpsum, Redemption) over time. Demographic breakdowns include Age Group Distribution (bar chart), Gender Split (pie chart), and State-wise Investor Map. The T30/B30 analysis compares folio density against AUM contribution across city tiers.



14.5 Page 5 — SIP & Market Trends

The SIP & Market Trends page integrates retail investment behavior with market index movements. A dual-axis chart overlays monthly SIP Inflows against the Nifty 50 index trend, highlighting how market rallies and corrections influence SIP investor behavior. Category Inflow trend lines reveal sector rotation patterns. Month-over-month SIP growth KPIs are prominently displayed.



15. Investor Analytics

The Investor Analytics module was developed to understand investor behavior, retention patterns, and SIP contribution continuity. Using the investor transaction dataset, cohort analysis and SIP continuity analysis were performed to identify engagement trends and potential risk indicators.

15.1 Cohort Analysis

Investors were grouped into monthly acquisition cohorts based on their first transaction date. For each cohort, the retention rate — defined as the proportion of investors who continued transacting in subsequent months — was computed. The cohort analysis grouped investors according to their first transaction period and tracked subsequent activity patterns. The analysis provided insights into investor engagement and long-term participation trends.

15.2 SIP Continuity Analysis

SIP investors were classified by their contribution pattern — Regular (monthly contributions with no gap), Irregular (gaps of 1–2 months), and At-Risk (gaps exceeding 3 months). The analysis identified the proportion of at-risk SIP investors by fund category and age group. Younger investors (25–30 years) showed the highest discontinuation rates, suggesting the need for automated reminders and financial counselling interventions.

The SIP continuity analysis classified 1,362 eligible investors based on contribution regularity. Among them, 1,332 investors were categorized as At Risk due to extended contribution gaps, while 30 investors were classified as Healthy. These findings can support investor retention and engagement strategies.

Enter Risk Profile (Low/Moderate/High): low

amfi_code	sharpe_ratio	risk_grade
119551	1.208	Low
119552	0.953	Low
102885	0.817	Low

SAMPLE OUTPUT

16. Recommendation Engine

A rule-based Fund Recommendation Engine was implemented in `recommender.py`, designed to match investors to fund categories based on their self-declared or inferred risk profile. The engine classifies investors into three risk tiers and recommends appropriate fund categories accordingly:

Risk Profile	Recommended Fund Categories	Rationale
Low Risk	Liquid Funds, Overnight Funds, Short Duration Debt Funds	Capital preservation, stable returns, short investment horizon
Moderate Risk	Hybrid Funds, Balanced Advantage Funds, Index Funds	Balanced growth, 3–5 year horizon, moderate market exposure
High Risk	Large Cap Equity, Mid Cap, Multi Cap, Small Cap Funds	Long-term wealth creation, 5+ year horizon, high return potential

In addition to risk profiling, the engine incorporates the fund's HHI score (to avoid highly concentrated funds for conservative investors), rolling Sharpe Ratio stability (to avoid funds with erratic risk-adjusted returns), and Maximum Drawdown thresholds (to filter funds unsuitable for low-risk investors). The engine outputs a ranked list of recommended schemes for each risk profile, exported as a recommendation report.

17. Key Results and Findings

The Mutual Fund Analytics Platform yielded significant analytical insights across performance, risk, and investor behavior dimensions. The key findings are summarized below:

17.1 Performance Findings

The fund performance analytics module provided a comprehensive assessment of mutual fund returns and risk-adjusted performance across multiple schemes. Metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were computed and consolidated into a unified fund scorecard for comparative analysis.

The analysis revealed significant variation in performance across schemes, highlighting the importance of evaluating both absolute returns and risk-adjusted returns. While several funds demonstrated strong return generation, the Sharpe Ratio and Sortino Ratio helped identify schemes that delivered more efficient returns relative to the level of risk undertaken. Funds with higher risk-adjusted performance consistently ranked above schemes that relied solely on high returns.

Alpha and Beta analysis provided additional insights into fund behavior relative to market benchmarks. Positive Alpha values indicated funds that outperformed benchmark expectations, while Beta values helped assess sensitivity to market movements. Maximum Drawdown analysis further highlighted the potential downside risk associated with each scheme, enabling a balanced evaluation of return potential and capital preservation.

Overall, the performance analytics framework demonstrated that a combination of return, volatility, benchmark comparison, and drawdown metrics provides a more reliable basis for mutual fund evaluation than relying on returns alone. These findings formed the foundation of the fund ranking, risk grading, and recommendation modules developed in the project.

17.2 Risk Findings

Advanced risk analytics provided deeper insights into downside risk and portfolio concentration. VaR and CVaR metrics were computed for 40 mutual fund schemes to estimate potential losses under adverse market conditions. Rolling Sharpe Ratio analysis across 5,750 observations helped evaluate the consistency of risk-adjusted returns over time. Additionally, HHI analysis of 34 portfolios identified varying levels of sector concentration, enabling assessment of diversification risk and portfolio stability.

17.3 Investor Behavior Findings

Investor analytics revealed meaningful patterns in participation and contribution behavior. SIP continuity analysis evaluated 1,362 eligible investors, of which 1,332 were classified as At Risk and 30 as Healthy based on contribution regularity. Cohort analysis provided insights into investor engagement trends and retention behavior over time. These findings can help financial institutions identify potential discontinuation risks and design targeted investor engagement strategies.

Metric	Key Finding
Top CAGR (5Y)	Small Cap Category: ~22% average; Large Cap Category: ~14% average

Best Sharpe Ratio	Index funds and select hybrid funds consistently lead in risk-adjusted returns
Highest Alpha	Active mid-cap and flexi-cap funds in the top quartile showed alpha of 3–5%
Highest VaR	Small Cap equity funds: daily VaR at 95% = -2.2% to -2.5%
Lowest VaR	Liquid and Overnight Debt Funds: daily VaR < -0.05%
At-Risk SIP Investors	~35% show contribution gaps within 6 months of SIP initiation
Industry AUM Concentration	Top 5 fund houses control >60% of total industry AUM
HHI Alert Funds	Sectoral/Thematic funds: HHI > 0.30 flagged as high concentration

18. Business Impact

The Mutual Fund Analytics Platform delivers practical business value by transforming fragmented mutual fund data into actionable insights for decision-making. The platform enables efficient performance evaluation, risk assessment, investor behavior analysis, and fund recommendation through a unified analytics framework.

18.1 Investor Intelligence

The platform provides insights into investor participation, retention, and SIP continuity patterns. These insights can help identify at-risk investors and support targeted engagement strategies.

18.2 Fund Selection Support

The Fund Scorecard and Recommendation Engine assist in evaluating and ranking mutual fund schemes using both return and risk-adjusted performance metrics, enabling more informed investment decisions.

18.3 Regulatory & Compliance Readiness

The structured SQLite database and documented data dictionary improve data consistency, transparency, and auditability, supporting reporting and governance requirements.

18.4 Scalability Foundation

The modular architecture allows future expansion of data sources, analytics models, and dashboard capabilities, making the platform suitable for further development and deployment.

19. Challenges Faced

The development of this platform encountered several real-world data engineering and analytics challenges:

- **Data Availability & Completeness:** AMFI and MFAPI do not always provide complete historical data for all schemes, particularly for recently launched funds or discontinued schemes. Custom imputation and flagging logic was required.
- **API Rate Limiting:** The MFAPI endpoint imposed implicit rate limits during batch NAV fetching. A sleep-and-retry mechanism with exponential backoff was implemented to handle request throttling gracefully.
- **NAV Data Gaps:** Non-trading day NAV gaps created discontinuities in return calculations. Forward-filling was applied consistently, though this introduces a minor assumption of return stationarity on non-trading days.
- **Benchmark Alignment:** Aligning fund-level NAV return dates with benchmark index return dates required careful date indexing and handling of market holidays that differ between fund and index calculation methodologies.
- **Schema Design Trade-offs:** Balancing normalization depth with query performance in SQLite required iterative schema refinement, particularly for the `fact_nav` and `fact_transactions` tables with high row counts.
- **Power BI SQLite Connectivity:** Establishing a stable ODBC connection between Power BI Desktop and SQLite on Windows required the installation and configuration of a compatible SQLite ODBC driver.

20. Limitations

While the platform delivers comprehensive analytics across multiple dimensions, several limitations should be acknowledged:

- **Simulated Transaction Data:** A portion of the investor transaction dataset is synthetic/simulated due to the non-availability of real investor-level transaction data in public domains. Findings from the investor analytics module should be interpreted as indicative rather than definitive for production use.
- **SQLite Scalability:** SQLite is appropriate for prototype and development workloads but is not designed for concurrent multi-user access or large-scale production deployments. Migration to PostgreSQL or a cloud data warehouse would be required for production scaling.
- **Static Recommendation Engine:** The current recommendation engine uses rule-based classification without machine learning. It does not adapt to changing market conditions, investor behavioral shifts, or emerging fund performance dynamics in real time.
- **Point-in-Time Benchmark Data:** Benchmark index data was collected at a specific point in time and does not auto-refresh. Real-time benchmark alignment requires integration with NSE/BSE live data feeds.
- **Limited Fund Universe:** The analytics cover actively managed and index funds registered with AMFI. Alternative investments such as Portfolio Management Services (PMS), Alternative Investment Funds (AIFs), and offshore funds are outside the scope.

21. Future Enhancements

The platform provides a strong foundation for several high-impact future enhancements:

- **Cloud Migration:** Migrate the SQLite database and Python pipeline to AWS (S3 + RDS/Redshift) or Azure (ADLS + Synapse Analytics) for production-grade scalability, reliability, and concurrent access.
- **Machine Learning Models:** Implement ML-based fund return prediction models (LSTM for NAV forecasting, clustering for fund peer grouping) and a collaborative filtering-based recommendation engine that personalizes suggestions based on investor transaction history.
- **Real-Time Data Streaming:** Integrate Apache Kafka or AWS Kinesis to enable real-time NAV updates and live risk metric computation, replacing the current batch ingestion approach.
- **Mobile Dashboard:** Extend the Power BI dashboard to a mobile-optimized format or build a React Native / Flutter application providing investors with on-the-go fund analytics and portfolio insights.
- **NLP-Based News Sentiment:** Incorporate news sentiment analysis using NLP models to overlay fund-specific news sentiment on NAV trend charts, providing contextual risk alerts.
- **ESG Integration:** Add Environmental, Social, and Governance (ESG) scoring for fund portfolios based on holdings data, aligning the platform with growing investor demand for responsible investing analytics.

22. Conclusion

The Mutual Fund Analytics Platform represents a comprehensive, production-ready analytics solution for the Indian mutual fund industry, built entirely using open-source Python technologies and publicly available regulatory data. Over six intensive project days, the platform successfully delivered ten integrated datasets, a normalized SQLite relational database, a full suite of financial performance and risk metrics, an investor analytics module, a rule-based recommendation engine, and a five-page interactive Power BI dashboard.

The project demonstrates that a rigorous, data-driven approach to mutual fund analysis — encompassing CAGR, Sharpe Ratio, Alpha/Beta, Maximum Drawdown, VaR, CVaR, HHI, and cohort analysis — provides significantly richer investment intelligence than traditional point-in-time return comparisons. The investor analytics findings, in particular, highlight actionable opportunities for Bluestock Fintech to improve SIP retention and investor engagement through targeted, data-driven interventions.

From a technical standpoint, the modular pipeline architecture ensures that each layer — ingestion, cleaning, storage, analytics, and visualization — can be independently extended or replaced as the platform scales. The project provides a strong foundation for Bluestock Fintech Pvt. Ltd. to build a market-leading mutual fund intelligence product that serves investors, advisors, and regulators alike.

This capstone experience has provided deep practical exposure to the intersection of data engineering, financial analytics, and business intelligence in a real-world fintech context — skills that are directly applicable to professional roles in the Indian fintech ecosystem.

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— End of Report —

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